CRYSTAL CHENG

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EDUCATION

University of California, Berkeley, Berkeley, CA — Expected Dec 2026

Bachelor of Arts in Computer Science & Applied Mathematics

- Coursework: Artificial Intelligence, Machine Learning, Data Structures, Statistical Modeling, Linear Algebra, Probability Theory
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer)

WORK EXPERIENCE

Hyve Solutions | Software Engineer Intern — June 2026 - Aug 2026

- Resolved an Azure OpenAI token-limit failure blocking concurrent users by replacing full-catalog retrieval with a Hugging Face embeddings + Neo4j vector-search pipeline, cutting per-request candidates **~95%** via a top-150 cap with a keyword fallback.
- Built a full-stack AI test-case search endpoint combining tiered keyword matching, RapidFuzz fuzzy prefiltering, and structured LLM re-ranking constrained to an ID allowlist to **eliminate hallucinated results**, reused by Analyze's backend to unify retrieval.
- Eliminated duplicate case-detail code across two navigation surfaces by building a single renderer that switches context via a page-state flag, letting the test-plan and test-case managers share one detail view with zero logic duplication.

TrackFly | Software Engineer Intern — Sep 2025 - Dec 2025

- Reduced cross-team integration mismatches on a **15-person team** by designing REST API contracts and MySQL schemas that defined a single source of truth for feature data across services.
- Cut per-feature build time by shipping 15+ reusable React/TypeScript components that standardized UI patterns across 10 weeks.

Barobo | Software Engineer Intern — May 2025 - Aug 2025

- **Eliminated unauthorized video sharing** across all user traffic by engineering single-use AWS CloudFront signed URLs with per-user access control and configurable TTL expiration.
- Scaled to **10,000+ monthly authorized video requests** by tracking user access metadata in AWS DynamoDB, enabling reliable per-request authorization without adding infrastructure overhead.
- Cut environment provisioning from **2 hours to 15 minutes** by rewriting deployment as reusable Terraform and AWS CLI scripts, making infrastructure reproducible across environments.

Opal Mentorship App | Machine Learning Intern — May 2024 - May 2025

- Raised mentor-mentee match accuracy **12**% across cohorts by calibrating 6 scoring weights over a 2-round data-backed process.
- Built a reproducible ML evaluation pipeline in AWS Lambda and DynamoDB that scores compatibility across 4 behavioral dimensions, giving the team **versioned, auditable weighting decisions** at cohort scale.

PROJECTS

RepoRank Github Search Engine | *Python, FastAPI, SQLAlchemy* — *link*

- Served BM25/BM25F search over 157k GitHub repos on a from-scratch field-aware inverted index: **p95 23.8ms** uncached, sub-millisecond median cached (**94.3% hit rate**), single-process throughput saturating at ~100 QPS (GIL-bound).
- Built an **offline ranking-eval gate** (nDCG/MRR/P@k with bootstrap confidence intervals over hand-labeled queries) wired into CI against a frozen 157k index, failing the build when nDCG regresses past a set margin.
- Crawled and indexed **157k repos** with a resumable rate-limited GitHub crawler, rebuilding a field-aware inverted index snapshot

FreshCheck Fruit Freshness Classifier | *PyTorch, ONNX, React, Vercel* — *link*

- Ran **~14ms** fruit-freshness inference fully in-browser with zero backend by training a 24K-parameter CNN in PyTorch and exporting to ONNX (WebAssembly), keeping all images on-device.
- Caught and fixed train/test leakage from duplicate source images by re-splitting on content and perceptual hashes into a leak-free held-out set, reporting honest **97.7% test accuracy (43/44)** with a 95% confidence interval.

MBTI Guesser | *Hugging Face Transformers BART-MNLI, DeepFace, OpenCV, Python* — *link*

- Built MBTI Guesser (Hugging Face Spaces), a multi-signal personality classifier inferring all four MBTI axes via zero-shot BART-MNLI, and validated the text pipeline on a hand-written unambiguous test set (**16/16 correct, >97% mean confidence**).
- Late-fused photo (DeepFace + OpenCV) and numeric signals with the text classification using weighted blending that **redistributes signal weight to text when photo input is unavailable**.
- **Diagnosed identity-framing bias** in the zero-shot NLI hypothesis template that skewed predictions, rewrote it around evidence-framing language, and added a confidence-gap threshold so ambiguous axes abstain instead of forcing a type.

SKILLS

- Languages: Python, JavaScript, TypeScript, C/C++, SQL, Bash, HTML/CSS (Tailwind)
- ML & AI: PyTorch, TensorFlow, Hugging Face Transformers, ONNX, RAG, Scikit-learn, NumPy, Pandas, OpenCV, DeepFace
- Backend & Data: FastAPI, Node.js, REST APIs, SQLAlchemy, PostgreSQL, DynamoDB, AWS (S3, CloudFront, Lambda)
- Frontend & Systems: React, Next.js, Docker, CI/CD (GitHub Actions, Jenkins), Terraform, Git, Linux, pytest, Vercel